

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2022 Worked Solutions

January 2022 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 January 2022

January 2022 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. The table below shows corresponding values of x and y for

$$y = 2^{5-\sqrt{x}}$$

The values of y are given to 3 decimal places.

x	5	5.5	6	6.5	7
y	6.792	6.298	5.858	5.466	5.113

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for

$$\int_5^7 2^{5-\sqrt{x}} dx$$

giving your answer to 2 decimal places.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

(i) $\int_5^7 2^{6-\sqrt{x}} dx$

(ii) $\int_5^7 (3 + 2^{5-\sqrt{x}}) dx$

(4)

Worked Solution - Question 1

1. Identify the strip width

The x values increase by 0.5, so $h = 0.5$.

2. Apply the trapezium rule

$$\int_5^7 2^{5-\sqrt{x}} dx \approx \frac{0.5}{2} \{6.792 + 5.113 + 2(6.298 + 5.858 + 5.466)\}.$$

3. Calculate the estimate

This gives 11.78725, so the estimate is 11.79 to 2 decimal places.

4. Use the factor of two

$$2^{6-\sqrt{x}} = 2 \cdot 2^{5-\sqrt{x}}, \text{ so } \int_5^7 2^{6-\sqrt{x}} dx \approx 2(11.78725) = 23.5745.$$

5. Estimate part i

This is 23.57 to 2 decimal places.

6. Add the constant integral

$$\int_5^7 (3 + 2^{5-\sqrt{x}}) dx = \int_5^7 3 dx + \int_5^7 2^{5-\sqrt{x}} dx.$$

7. Estimate part ii

The estimate is $3(7 - 5) + 11.78725 = 17.78725$, so 17.79 to 2 decimal places.

Final answer

(a) 11.79. (b)(i) 23.57. (b)(ii) 17.79.

Question 2

Applications of Differentiation

2. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving each term in simplest form. (2)
- (b) Hence find the coordinates of the stationary point of C . (4)
- (c) Find $\frac{d^2y}{dx^2}$ and hence determine the nature of the stationary point of C . (2)

Worked Solution - Question 2

Topic group

1. Differentiate

For $y = 27x^{1/2} - x^{3/2} - 20$, $\frac{dy}{dx} = \frac{27}{2}x^{-1/2} - \frac{3}{2}x^{1/2}$.

2. Set the gradient to zero

$$\frac{27}{2}x^{-1/2} - \frac{3}{2}x^{1/2} = 0.$$

3. Clear the powers

Multiplying by $2x^{1/2}$ gives $27 - 3x = 0$, so $x = 9$.

4. Find y

At $x = 9$, $y = 27\sqrt{9} - 9^{3/2} - 20 = 81 - 27 - 20 = 34$.

5. State the stationary point

The stationary point is $(9, 34)$.

6. Differentiate again

$$\frac{d^2y}{dx^2} = -\frac{27}{4}x^{-3/2} - \frac{3}{4}x^{-1/2}.$$

7. Determine the nature

At $x = 9$, both terms in $\frac{d^2y}{dx^2}$ are negative, so the stationary point is a local maximum.

Final answer

(a) $\frac{dy}{dx} = \frac{27}{2}x^{-1/2} - \frac{3}{2}x^{1/2}$. (b) $(9, 34)$. (c) local maximum.

Question 3

Binomial Expansion

3. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{kx}{4}\right)^8$$

where k is a non-zero constant. Give each term in simplest form.

(4)

$$f(x) = (5 - 3x)\left(2 - \frac{kx}{4}\right)^8$$

In the expansion of $f(x)$, the constant term is 3 times the coefficient of x .

- (b) Find the value of k .

(3)

Worked Solution - Question 3

1. Write the first terms

$$\left(2 - \frac{kx}{4}\right)^8 = 2^8 + \binom{8}{1}2^7\left(-\frac{kx}{4}\right) + \binom{8}{2}2^6\left(-\frac{kx}{4}\right)^2 + \binom{8}{3}2^5\left(-\frac{kx}{4}\right)^3 + \dots$$

2. Simplify the expansion

The first four terms are $256 - 256kx + 112k^2x^2 - 28k^3x^3$.

3. Find the constant term in f

For $f(x) = (5 - 3x)\left(2 - \frac{kx}{4}\right)^8$, the constant term is $5(256) = 1280$.

4. Find the coefficient of x

The coefficient of x is $5(-256k) - 3(256) = -1280k - 768$.

5. Use the condition

The constant term is 3 times the coefficient of x , so $1280 = 3(-1280k - 768)$.

6. Solve for k

$$1280 = -3840k - 2304, \text{ hence } k = -\frac{3584}{3840} = -\frac{14}{15}.$$

Final answer

$$(a) 256 - 256kx + 112k^2x^2 - 28k^3x^3. \quad (b) k = -\frac{14}{15}.$$

Question 4

Laws of Logarithms

4. Using the laws of logarithms, solve

$$\log_3(32 - 12x) = 2\log_3(1 - x) + 3$$

(5)

Worked Solution - Question 4

1. Rewrite the right side

$$2\log_3(1-x) = \log_3(1-x)^2 \text{ and } 3 = \log_3 27.$$

2. Combine the logarithms

The equation becomes $\log_3(32 - 12x) = \log_3\{27(1-x)^2\}$.

3. Equate the arguments

$$32 - 12x = 27(1-x)^2.$$

4. Form the quadratic

$$32 - 12x = 27 - 54x + 27x^2, \text{ so } 27x^2 - 42x - 5 = 0.$$

5. Solve

$$x = \frac{42 \pm 48}{54}, \text{ giving } x = \frac{5}{3} \text{ or } x = -\frac{1}{9}.$$

6. Check the domain

The original logarithms require $1-x > 0$, so $x < 1$. Therefore $x = \frac{5}{3}$ is invalid and $x = -\frac{1}{9}$.

Final answer

$$x = -\frac{1}{9}.$$

Question 5

Polynomials

5. $f(x) = 3x^3 + Ax^2 + Bx - 10$

where A and B are integers.

Given that

- when $f(x)$ is divided by $(x - 1)$ the remainder is k
- when $f(x)$ is divided by $(x + 1)$ the remainder is $-10k$
- k is a constant

(a) show that

$$11A + 9B = 83 \quad (3)$$

Given also that $(3x - 2)$ is a factor of $f(x)$,

(b) find the value of A and the value of B . (3)

(c) Hence find the quadratic expression $g(x)$ such that

$$f(x) = (3x - 2)g(x) \quad (2)$$

Worked Solution - Question 5

1. Use x equals one

$$f(1) = 3 + A + B - 10 = A + B - 7, \text{ so } k = A + B - 7.$$

2. Use x equals negative one

$$f(-1) = -3 + A - B - 10 = A - B - 13.$$

3. Use the second remainder relation

$$\text{Since } f(-1) = -10k, A - B - 13 = -10(A + B - 7).$$

4. Show the relation

$$\text{Rearranging gives } 11A + 9B = 83.$$

5. Use the factor

$$\text{Since } (3x - 2) \text{ is a factor, } f\left(\frac{2}{3}\right) = 0.$$

6. Form the second equation

$$3\left(\frac{2}{3}\right)^3 + A\left(\frac{2}{3}\right)^2 + B\left(\frac{2}{3}\right) - 10 = 0. \text{ Multiplying by 9 gives } 2A + 3B = 41$$

7. Solve for A and B

$$\text{Solving } 11A + 9B = 83 \text{ with } 2A + 3B = 41 \text{ gives } A = -8 \text{ and } B = 19.$$

8. Divide by the factor

$$\text{With these values, } f(x) = 3x^3 - 8x^2 + 19x - 10 = (3x - 2)(x^2 - 2x + 5).$$

Final answer

(a) Proven: $11A + 9B = 83$.

(b) $A = -8$, $B = 19$. (c) $g(x) = x^2 - 2x + 5$.

Question 6

6.

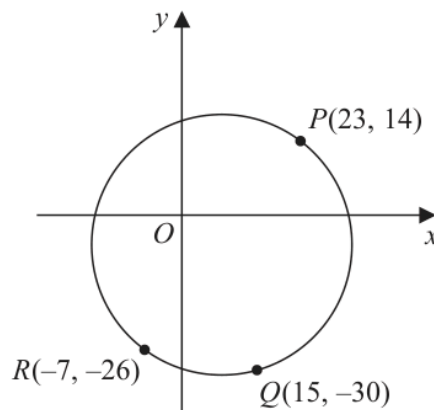


Figure 1

The points $P(23, 14)$, $Q(15, -30)$ and $R(-7, -26)$ lie on the circle C , as shown in Figure 1.

- (a) Show that $\angle PQR = 90^\circ$ (2)
- (b) Hence, or otherwise, find
- (i) the centre of C ,
 - (ii) the radius of C . (3)

Given that the point S lies on C such that the distance QS is greatest,

- (c) find an equation of the tangent to C at S , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (3)

Worked Solution - Question 6

1. Find the gradient QP

The gradient of QP is $\frac{14 - (-30)}{23 - 15} = \frac{44}{8} = \frac{11}{2}$.

2. Find the gradient QR

The gradient of QR is $\frac{-26 - (-30)}{-7 - 15} = \frac{4}{-22} = -\frac{2}{11}$.

3. Show the right angle

The product of the two gradients is -1 , so QP is perpendicular to QR and $\angle PQR = 90^\circ$.

4. Use the diameter

Since the angle at Q is a right angle, PR is a diameter of the circle.

5. Find the centre

The centre is the midpoint of $P(23, 14)$ and $R(-7, -26)$, so it is $(8, -6)$.

6. Find the radius

$PR = \sqrt{30^2 + 40^2} = 50$, so the radius is 25 .

7. Find the farthest point from Q

The greatest distance from Q occurs at the point opposite Q on the diameter through the centre. Thus $S = 2(8, -6) - (15, -30) = (1, 18)$.

8. Find the tangent gradient

The radius from the centre to S has gradient $\frac{18 - (-6)}{1 - 8} = -\frac{24}{7}$, so the tangent gradient is $\frac{7}{24}$.

9. Write the tangent equation

$$y - 18 = \frac{7}{24}(x - 1), \text{ which rearranges to } 7x - 24y + 425 = 0.$$

Final answer

(a) Proven: $\angle PQR = 90^\circ$.

(b)(i) $(8, -6)$. (b)(ii) 25. (c) $7x - 24y + 425 = 0$.

Question 7

Trigonometric Equations

7. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-90^\circ < x < 90^\circ$, the equation

$$3 \sin(2x - 15^\circ) = \cos(2x - 15^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) Solve, for $0 < \theta < 2\pi$, the equation

$$4 \sin^2 \theta + 8 \cos \theta = 3$$

giving your answers to 3 significant figures.

(4)

Worked Solution - Question 7

1. Solve the first tangent equation

Let $u = 2x - 15^\circ$. Then $3 \sin u = \cos u$, so $\tan u = \frac{1}{3}$.

2. Use the interval for u

Since $-90^\circ < x < 90^\circ$, $-195^\circ < u < 165^\circ$.

3. Find x

$u = 18.434\dots^\circ$ or $u = -161.565\dots^\circ$, so $x = 16.7^\circ$ or $x = -73.3^\circ$.

4. Rewrite the second equation

$$4 \sin^2 \theta + 8 \cos \theta = 3.$$

5. Use sin squared

Substitute $\sin^2 \theta = 1 - \cos^2 \theta$ to get $4(1 - \cos^2 \theta) + 8 \cos \theta = 3$.

6. Solve for cos theta

This gives $4 \cos^2 \theta - 8 \cos \theta - 1 = 0$, so $\cos \theta = 1 \pm \frac{\sqrt{5}}{2}$.

7. Choose the valid cosine

Only $\cos \theta = 1 - \frac{\sqrt{5}}{2}$ lies between -1 and 1 .

8. State theta

For $0 < \theta < 2\pi$, $\theta = 1.69$ or 4.59 to 3 significant figures.

Final answer

(i) $x = -73.3^\circ, 16.7^\circ$. (ii) $\theta = 1.69, 4.59$.

Question 8

Modelling with Sequences & Series

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

(a) show that the post is driven an **additional** 62 mm into the ground with the 20th hit. (1)

(b) Find the **total distance** that the post has been driven into the ground after 20 hits. (2)

Given that for each subsequent hit after the 20th hit

- the **additional** distances the post travels form a geometric sequence with common ratio r
- on the 22nd hit, the post is driven an **additional** 60 mm into the ground

(c) find the value of r , giving your answer to 3 decimal places. (2)

After a total of N hits, the post will have been driven more than 3 m into the ground.

(d) Find, showing all steps in your working, the smallest possible value of N . (4)

Worked Solution - Question 8

Topic group

1. Use the arithmetic rule

For the first **20** hits, the additional distances form an arithmetic sequence with first term **100** and common difference **-2**.

2. Show the twentieth hit

The twentieth additional distance is $100 + 19(-2) = 62$ mm.

3. Sum the first twenty hits

$$S_{20} = \frac{20}{2}(100 + 62) = 1620 \text{ mm.}$$

4. Use the geometric information

After the twentieth hit, the distances follow the same ratio, so the twenty second distance is $62r^2$.

5. Find r

$$62r^2 = 60, \text{ hence } r = \sqrt{\frac{30}{31}} = 0.984 \text{ to 3 decimal places.}$$

6. Set up the remaining distance

More than **3** m means more than **3000** mm. After **20** hits, the remaining amount to exceed is $3000 - 1620 = 1380$ mm.

7. Sum the geometric tail

$$\text{If there are } m \text{ further hits, their total is } 62r + 62r^2 + \dots + 62r^m = 62r \frac{1 - r^m}{1 - r}.$$

8. Find the first m that works

Using $r = \sqrt{30/31}$, $62r \frac{1 - r^{27}}{1 - r}$ is not enough, but $62r \frac{1 - r^{28}}{1 - r}$ makes the total exceed 3000 mm.

9. State N

Therefore $m = 28$ further hits are needed after the first 20, so
 $N = 20 + 28 = 48$.

Final answer

(a) Proven: 62 mm.

(b) 1620 mm.

(c) $r = 0.984$. (d) $N = 48$.

Question 9

Integration

9.

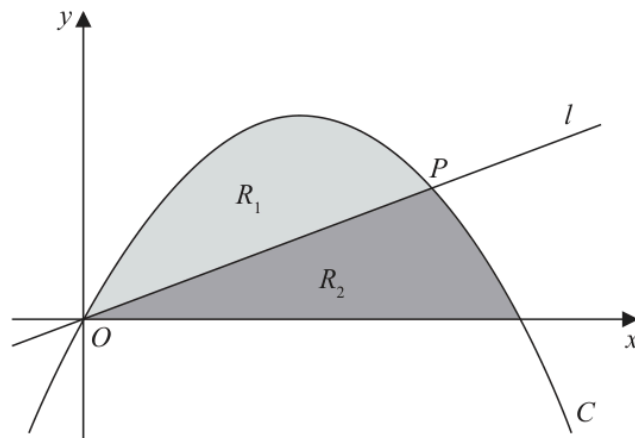


Figure 2

Figure 2 shows

- the curve C with equation $y = x - x^2$
- the line l with equation $y = mx$, where m is a constant and $0 < m < 1$

The line and the curve intersect at the origin O and at the point P .

- (a) Find, in terms of m , the coordinates of P .

(2)

The region R_1 , shown shaded in Figure 2, is bounded by C and l .

- (b) Show that the area of R_1 is

$$\frac{(1-m)^3}{6}$$

(5)

The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and l .

Given that the area of R_1 is equal to the area of R_2

- (c) find the exact value of m .

(3)

Worked Solution - Question 9

1. Find the second intersection

At an intersection, $x - x^2 = mx$, so $x(1 - m - x) = 0$.

2. State P

The non-zero intersection has $x = 1 - m$, and then $y = m(1 - m)$, so $P = (1 - m, m(1 - m))$.

3. Set up area R one

Region R_1 is between the curve and the line from $x = 0$ to $x = 1 - m$.

4. Integrate the difference

$$A_{R_1} = \int_0^{1-m} \{(x - x^2) - mx\} dx = \int_0^{1-m} \{(1 - m)x - x^2\} dx.$$

5. Evaluate

$$A_{R_1} = \left[\frac{1-m}{2} x^2 - \frac{1}{3} x^3 \right]_0^{1-m} = \frac{(1-m)^3}{2} - \frac{(1-m)^3}{3} = \frac{(1-m)^3}{6}.$$

6. Use the total curve area

The total area under $y = x - x^2$ from $x = 0$ to $x = 1$ is $\int_0^1 (x - x^2) dx = \frac{1}{6}$.

7. Use equal regions

Since R_1 and R_2 are equal and together make this total area, each has area $\frac{1}{12}$.

8. Solve for m

$$\frac{(1-m)^3}{6} = \frac{1}{12}, \text{ so } (1-m)^3 = \frac{1}{2} \text{ and } m = 1 - \sqrt[3]{\frac{1}{2}}.$$

Final answer

(a) $P = (1 - m, m(1 - m))$. (b) Proven: $\frac{(1 - m)^3}{6}$.

(c) $m = 1 - \frac{1}{\sqrt[3]{2}}$.

Question 10

Proof

10. (i) Prove by counter example that the statement

“if p is a prime number then $2p + 1$ is also a prime number”

is not true.

(1)

(ii) Use proof by exhaustion to prove that if n is an integer then

$$5n^2 + n + 12$$

is always even.

(4)

(Total 5 marks)

Worked Solution - Question 10

Topic group

1. Give a counterexample

Take $p = 7$, which is prime.

2. Show the statement fails

$2p + 1 = 2(7) + 1 = 15$, and 15 is not prime. So the statement is false.

3. Case one: n even

If n is even, let $n = 2k$. Then

$5n^2 + n + 12 = 5(2k)^2 + 2k + 12 = 20k^2 + 2k + 12 = 2(10k^2 + k + 6)$,
which is even.

4. Case two: n odd

If n is odd, let $n = 2k + 1$. Then $5n^2 + n + 12 = 5(2k + 1)^2 + (2k + 1) + 12$.

5. Simplify the odd case

$5(2k + 1)^2 + (2k + 1) + 12 = 20k^2 + 22k + 18 = 2(10k^2 + 11k + 9)$, which
is even.

6. Conclude by exhaustion

Every integer is either even or odd, and both cases give an even result. Therefore
 $5n^2 + n + 12$ is always even.

Final answer

(i) For $p = 7$, $2p + 1 = 15$, not prime.

(ii) Proven.